

LODZ, Poland

WMO: 124650

Lat: 51.723N

Lon: 19.399E

Elev: 190

StdP: 99.06

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-11.8	-17.4	0.8	-13.9	-14.7	1.1	-10.8	11.2	4.1	10.2	3.2	2.3	100	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	30.3	19.9	28.3	19.0	26.7	18.3	21.1	27.4	20.2	26.1	19.3	24.6	3.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.2	23.1	18.2	13.4	22.2	17.4	12.7	21.4	62.0	27.3	58.9	26.2	55.8	24.5	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.0	7.0	-18.1	33.6	4.3	1.8	-21.2	34.9	-23.7	36.0	-26.1	37.0	-29.1	38.3		
(5)			-18.4	22.7	4.1	0.9	-21.4	23.4	-23.8	24.0	-26.2	24.5	-29.2	25.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-1.8	-0.4	3.1	8.9	13.7	16.8	19.1	18.7	13.8	8.8	4.0	-0.1
	DBStd	8.47	5.35	4.95	4.41	4.37	3.97	3.57	3.42	3.26	3.48	4.11	4.15	4.89
	HDD10.0	1527	367	291	217	70	11	0	0	0	6	72	182	312
	HDD18.3	3632	625	524	473	284	153	70	33	36	141	294	429	570
	CDD10.0	1078	0	0	2	37	124	205	281	269	121	37	3	0
	CDD18.3	141	0	0	0	1	8	25	56	47	5	0	0	0
	CDH23.3	1403	0	0	0	16	100	272	538	418	58	1	0	0
	CDH26.7	387	0	0	0	2	16	73	167	118	12	0	0	0
Wind	WSAvg	3.2	3.8	3.6	3.6	3.3	3.0	2.9	2.8	2.6	2.8	3.1	3.4	3.7
Precipitation	PrecAvg	579	35	33	38	36	63	65	85	55	48	39	43	39
	PrecMax	753	73	73	80	89	157	159	250	129	112	128	99	89
	PrecMin	406	5	4	15	1	21	12	22	17	13	6	0	8
	PrecStd	95	15	16	15	19	32	40	54	25	24	28	20	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	11.7	17.5	25.6	28.6	31.9	33.3	32.9	28.2	22.0	15.1	11.1
		MCWB	7.9	7.9	10.5	15.4	17.9	20.7	20.1	20.2	18.3	16.0	11.4	9.2
	2%	DB	7.9	9.5	14.5	22.0	26.1	28.8	30.5	29.7	24.8	19.5	13.0	9.0
		MCWB	6.4	7.1	8.9	13.5	17.0	19.2	19.8	19.9	16.9	14.7	10.8	7.5
	5%	DB	6.2	7.9	12.1	19.1	23.9	26.3	28.4	27.5	22.6	17.2	11.2	7.4
		MCWB	5.0	6.1	7.9	11.9	16.1	18.0	19.2	19.0	16.1	13.6	9.6	6.4
	10%	DB	4.9	6.1	9.9	16.8	21.5	24.2	26.4	25.5	20.5	15.0	9.9	6.0
		MCWB	3.9	4.5	6.9	10.8	14.8	17.0	18.5	18.0	15.2	12.5	8.5	5.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	8.8	11.2	16.2	19.2	21.7	22.5	22.2	19.4	17.0	12.6	9.6
		MCDB	9.4	10.3	16.1	23.8	26.1	28.7	28.9	28.3	25.6	20.5	14.4	10.7
	2%	WB	6.5	7.5	9.6	14.3	18.0	20.2	21.3	20.9	17.9	15.3	11.2	7.7
		MCDB	7.5	9.1	13.1	20.8	24.0	26.6	27.4	27.2	22.7	18.4	12.7	8.7
	5%	WB	5.2	6.1	8.4	12.8	16.8	19.0	20.4	19.9	16.9	14.1	9.9	6.4
		MCDB	6.2	7.6	11.3	17.7	22.8	24.5	26.0	25.9	21.3	16.7	11.1	7.3
	10%	WB	3.8	4.6	7.0	11.4	15.6	18.0	19.5	19.0	15.8	12.8	8.6	5.0
		MCDB	4.6	5.9	9.4	15.6	20.4	23.0	24.4	24.0	19.5	14.9	9.7	5.8
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.9	7.7	10.0	10.5	10.4	10.4	10.6	9.5	7.8	5.3	4.5
		MCDBR	5.3	7.9	11.9	14.3	14.0	14.1	13.9	13.7	13.3	11.1	8.0	5.4
		MCWBR	4.5	5.8	7.3	7.3	6.5	6.3	5.3	5.7	6.5	6.9	5.8	4.4
	5% WB	MCDBR	5.2	7.3	10.7	12.4	12.5	12.2	11.4	12.0	11.5	10.3	7.4	5.2
		MCWBR	4.5	5.7	7.1	6.8	6.3	6.1	5.1	5.5	6.2	6.6	5.7	4.5
Clear-Sky Solar Irradiance	taub	0.316	0.343	0.379	0.431	0.422	0.417	0.447	0.435	0.398	0.369	0.350	0.313	
	taud	2.260	2.260	2.216	2.127	2.206	2.253	2.182	2.219	2.289	2.341	2.319	2.301	
	Ebn at Noon	673	763	804	802	832	839	803	791	777	724	628	614	
	Edh at Noon	68	89	113	139	135	130	137	126	105	82	65	56	
All-Sky Solar Radiation	RadAvg	0.72	1.42	2.64	4.15	5.18	5.42	5.05	4.51	3.23	1.82	0.82	0.55	
	RadStd	0.14	0.25	0.32	0.55	0.54	0.52	0.60	0.38	0.45	0.28	0.11	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.51	N/A	N/A	N/A	+0.44	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		+0.51	N/A	N/A	N/A	N/A	N/A	N/A	-164	+78	+31

Nomenclature: See separate page